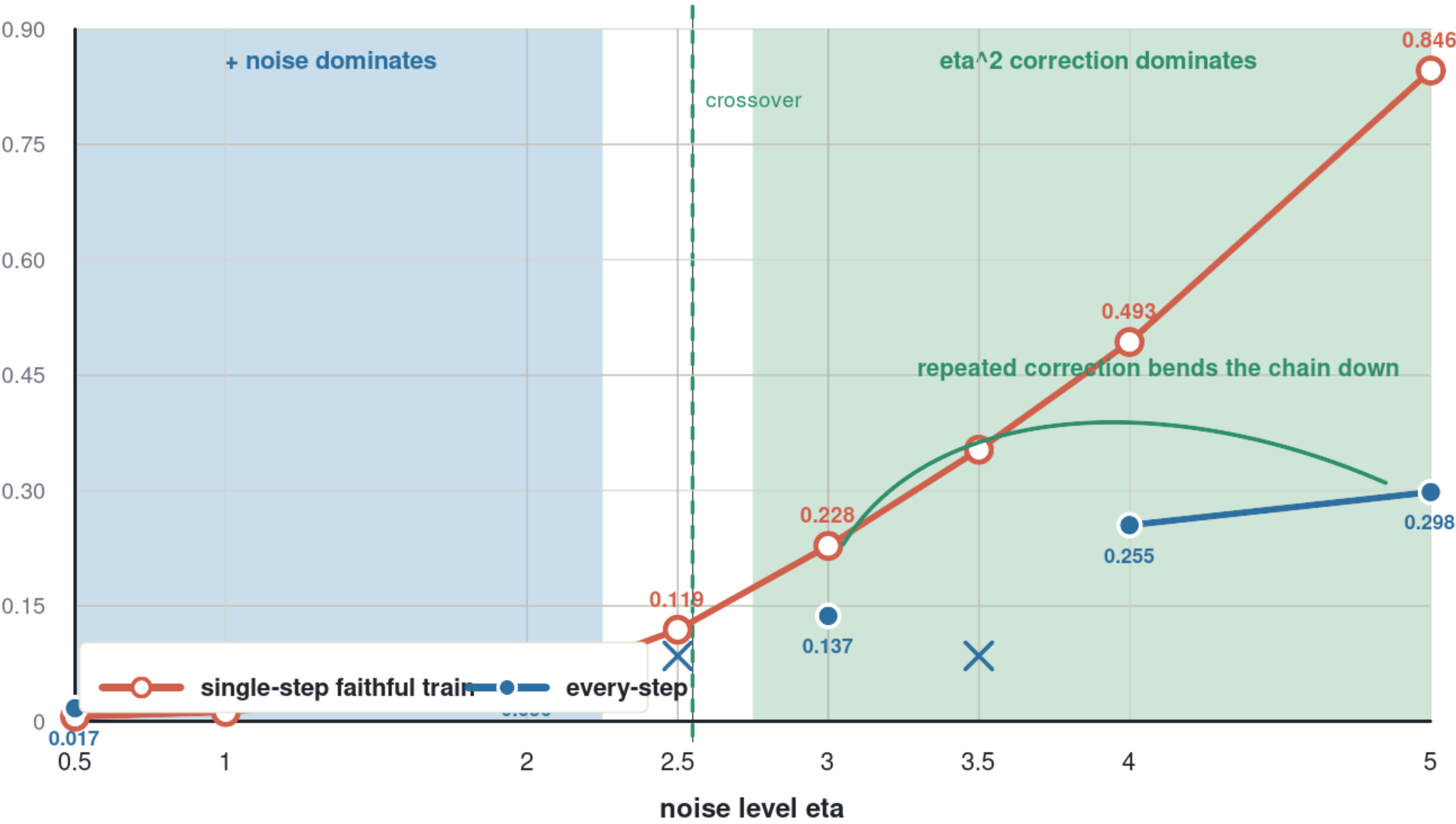


Flow-SDE noise injection

Why every-step sampling can end with lower action std than single-step sampling

Observed final action std

single stochastic denoise step vs stochastic transition at every denoise step



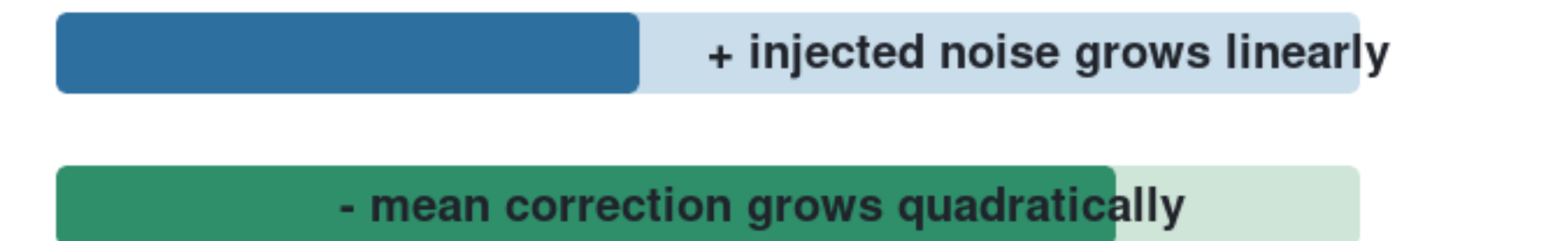
Mechanism in one transition

Flow-SDE is not just additive noise

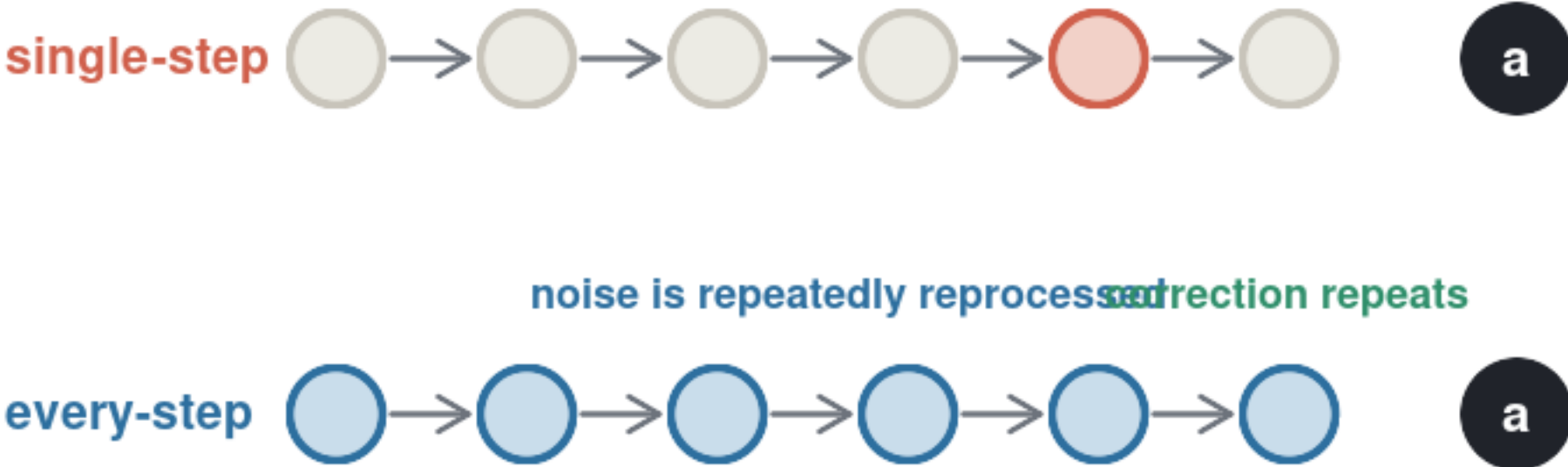
$$\begin{aligned} x_{\text{next}} &= \text{ODE}_{\text{next}} \\ &+ \alpha_i \cdot \text{eps} \end{aligned}$$

$$\begin{aligned} &- c_i \cdot \text{noise_pred} \\ \alpha_i &\propto \eta \quad c_i \propto \eta^2 \end{aligned}$$

competing terms



sampling paths



Takeaway

Every-step adds more stochastic transitions, but Flow-SDE also applies an η^2 mean correction at every transition. At large η , correction can dominate and lower the final action std.